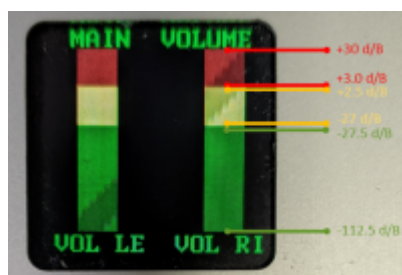


CONFON Headphones

In the 3T (Quattro & Penta) scanner suites audio is played via an Edirol [UA-4FX](#) USB soundcard.

[Confon](#) These are MRI-compatible headphones incorporated into ear mufflers for passive gradient noise reduction, signals are fed through the access panel via feed-through tubular filters together with filter/transformer-boxes. NB. These **ONLY** fit in the 20-Ch head coil (QUATTRO & PENTA).

Availability: [Quattro](#), [Penta](#),



SETTING THE PARTICIPANT VOLUME

With regards to the Confon system, the Confon headphones require the B0 magnetic field to work, consequently we do not have the required equipment with which to test the headphones at source.

Volume at the headphones is calculated by: Stimulus level + STIM-PC + Edirol Sound Card + CONFON amplifier = Headphone volume

The manufacturer has advised that we can expect a 116dB at the headphones with the following:

STIM-PC set to volume 100 (0dB) + Edirol set to maximum (-10dB) + CONFON amplifier set at 0dB = 116dB Note: this does not factor in your stimulus level.

The CONFON amplifier has a range of -115 dB to +30 dB (see attached picture).

If you were presenting stimuli at -20dB stimulus level (-20 dB) + STIM-PC set to volume 100 (0dB) + Edirol set to maximum (-10dB) + CONFON amplifier set at -27dB = 69dB

Therefore, this together with an example stimulus level of -20db, will result in an SPL of 126 -10 -27 -20-0 =69dB at the headphones

An example as to how this may be achieved, follows System amplification 126dB fixed stim pc - @ max volume 100 = (0dB) -fixed Edirol UA-4fx =-10dBu -fixed stimulus level (you set this) e.g. -20dB *user controlled amplifier volume setting e.g. -27dB *user controlled

To arrive at the sound level arriving at the Confon headphones the following formula maybe used.

System amplification - stimulus level- (PC level) - (amplifier setting) = SPL 126-10-20-0-27dB = 69dB

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